



Aircraft ordnance

Aircraft ordnance or **ordnance** (in the context of military aviation) is any expendable weaponry (e.g. bombs, missiles, rockets and gun ammunition) used by military aircraft. The term is often used when describing the payload of air-to-ground weaponry that can be carried by the aircraft or the weight that has been dropped in combat. Aircraft ordnance also includes air-to-air, anti-ship and anti-submarine weapons.

Some aircraft types can carry a wide variety of ordnance – for example, the Fairchild AU-23 Peacemaker could use forward-firing gun pods, 500 and 250 pound bombs, napalm units, cluster bomb units, flares, rockets, smoke grenades and propaganda leaflet dispensers.^[1]

Ordnance can be carried in a bomb bay or hung from a hardpoint.

For many weapons there is a limit to the length of time they can be flown (e.g. because of vibration damage); after this their safety or effectiveness is not guaranteed. This can be a problem if weapons designed for high intensity conflict are carried on multiple missions in a long counter-insurgency campaign.^[2]

History

Guidance

Ordnance can be divided into **guided** and **unguided** categories based on their targeting capabilities. **Guided ordnance**, such as Precision-guided munitions (*PGMs*), use advanced technologies like GPS, infrared, radar or laser tracking to precisely hit designated targets.^{[3][4][5]} **Unguided ordnance**, commonly known as "dumb bombs," lack guidance systems and rely on the aircraft's release parameters to strike a target. Though less accurate, unguided bombs are often used for area strikes and are more cost-effective. ^[5]



Grumman A-6 Intruder (A2F-1) attack aircraft displaying its weapons array, 1962



Air-to-Ground vs. Air-to-Air Ordnance

See also

- [List of aircraft weapons](#)

References

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2. De-classified summary of conference at UK [Joint Services Command and Staff College](#) reported in *Aerospace International* (magazine) March 2011 page 24
3. "Guest Editor's Introduction: Homing Missile Guidance and Control" (https://secwww.jhuapl.edu/techdigest/content/techdigest/pdf/V29-N01/29-01-Palumbo_GuestEditor.pdf) (PDF). *jhuapl.edu*.
4. "Laser-Guided Bombs - Building LGB Capability" (<https://www.sps-aviation.com/story/?id=1008>). *www.sps-aviation.com*. Retrieved 2024-11-16.
5. "Precision-guided munitions" (<https://www.baesystems.com/en-us/productfamily/precision-guided-munitions>). *www.baesystems.com*. Retrieved 2024-11-16.



Airman inspecting a [Joint Direct Attack Munition](#) (JDAM) after being loaded onto an [F-22 Raptor](#)



A crew member of a British [SS 'Z' Class airship](#) about to throw a bomb from the rear cockpit of the gondola.